

Presenter: Aishwarya Bhati

Roll Number: 2502221530017

Degree / Branch: B.Tech CSE (AI/ML), Section A

Institution: ITS Engineering College, Greater Noida (AKTU)

Target Duration: 5 Minutes 30 Seconds

Live Deployed App: <https://wellbridgeai.resence.in>



Word-for-Word Presentation Script (Slide-by-Slide)

Slide 1: Title & Academic Cover

0:40 mins

Stage Direction: Stand upright with a calm, confident smile. Make direct eye contact with the evaluators before speaking.

"Good morning respected evaluators, faculty members, and peers. My name is **Aishwarya Bhati**, Roll Number **2502221530017**, second-year B.Tech in Computer Science and Engineering with specialization in AI and Machine Learning, Section A, from **ITS Engineering College, Greater Noida**.

Today, I am proud to present **WellBridge AI** — an intelligent multimodal health companion engineered to bridge the communication gap between confusing medical diagnostic reports and everyday life, directly aligned with **United Nations Sustainable Development Goal 3: Good Health and Well-Being**."

→ [Cue: Click next slide button smoothly]

Slide 2: The Healthcare Dilemma (Problem Statement)

0:50 mins

Stage Direction: Use a serious, empathetic tone. Point gently towards the 3 problem cards on the screen.

"To understand why WellBridge AI is necessary, let us look at the current healthcare dilemma:

First, **The Cryptic Report Barrier**: Over 75% of patients receive diagnostic lab sheets filled with complex biomarker acronyms like ALT, eGFR, MCV, or HbA1c. Unable to decipher these numbers, patients turn to alarming internet searches that induce extreme health anxiety.

Second, **The 15-Minute Consultation Rush**: During doctor appointments, patients frequently forget critical symptom timelines or medication reactions experienced weeks prior. In fact, clinical studies show that patients forget **40 to 80% of medical advice** immediately after leaving the clinic.

Third, **Fragmented Medical Memory**: Patient records remain scattered across physical paper slips, WhatsApp photos, and unorganized folders, with zero unified chronological tracking."

→ [Cue: Transition to Slide 3]

Slide 3: The Solution — WellBridge AI Architecture

 0:45 mins

Stage Direction: *Speak with positive energy and clarity. Highlight the 4 distinct pillars.*

"To solve this, we developed **WellBridge AI** — an integrated full-stack platform built on four core clinical pillars:

- 1. Multimodal Lab Report Scanner:** Using computer vision to instantly parse blood panels and prescriptions.
- 2. Contextual AI Health Companion:** A profile-aware conversational engine providing personalized wellness explanations.
- 3. Daily Symptom & Mood Journal:** An encrypted daily reflection space that tracks symptom progression and mood biomarkers.
- 4. 1-Page Doctor Visit Brief Generator:** A one-click synthesis tool that compiles 30 days of health history into a standardized summary prepared for physicians."

→ [Cue: Transition to Slide 4]

Slide 4: Feature Deep-Dive — Lab Scanner & AI Companion

 0:50 mins

Stage Direction: Highlight the technical triage classifications (Normal, Monitor, Alert).

"Let's dive into the core intelligence layers:

Our **Multimodal Vision OCR Scanner** allows patients to upload a photo of any diagnostic panel. The AI extracts biomarker names, clinical units, and reference ranges, categorizing each metric into three clear triage levels: **Normal (Green), Monitor (Amber), or Alert (Red)**.

Simultaneously, the system automatically formulates **three targeted, high-value questions** for the patient to ask their doctor.

Furthermore, our **AI Health Companion** integrates the patient's baseline profile — including ongoing conditions, allergies, medications, and lifestyle. It operates under a **Strict Concise Clinical Protocol**, delivering high-signal, empathetic guidance under 150 words without generic robotic filler."

→ [Cue: Transition to Slide 5]

Slide 5: Continuous Wellness Tracking & Clinical Handoff

 0:50 mins

Stage Direction: Emphasize the unique 1-page printable doctor brief innovation.

"For ongoing care, WellBridge AI provides **Empathetic Daily Symptom Journaling**. Patients can simply type or voice how they feel each day, and the system automatically tags health metrics like fatigue, hydration, or medication response.

When it is time for a hospital visit, the patient clicks '**Generate Doctor Visit Brief**'.

Our engine compiles all symptom logs, medication reactions, and lab scan anomalies from the past 30 days into a structured, **one-page clinical brief**. We engineered a custom **Print Isolation Engine** so patients can print or save a clean, professional medical document formatted specifically for busy doctors to review in under 60 seconds."

→ [Cue: Transition to Slide 6]

Slide 6: Modern, Scalable & Secure Full-Stack Engineering

🕒 0:50 mins

Stage Direction: Use a confident technical tone. Evaluators love database security and AI fallback depth.

"Behind the scenes, WellBridge AI is engineered with production-grade full-stack standards:

- On the **Frontend**, we built a responsive Single Page Application using **React 18, TypeScript, Tailwind CSS**, and Progressive Web App capabilities for offline installation.
- On the **AI Intelligence Layer**, we utilize **Google Gemini Multimodal APIs** featuring a **4-tier dynamic fallback ladder** cascading from Flash-Lite to 3.7 Flash, guaranteeing zero downtime.
- On the **Backend & Database**, we utilize **Google Firebase Firestore** with strict row-level security rules enforcing that `request.auth.uid == userId`, ensuring mathematical zero cross-patient data exposure."

→ [Cue: Transition to Slide 7]

Slide 7: Transforming Patient Outcomes & Clinical Efficiency

🕒 0:35 mins

Stage Direction: Connect technology back to human impact and UN SDG 3.

"The impact of WellBridge AI is two-fold:

For **Patients**, it eliminates medical jargon anxiety and promotes proactive self-care.

For **Physicians**, it replaces vague, emotional memory with structured clinical timelines, dramatically increasing diagnostic efficiency during 15-minute consultations.

This directly aligns with **UN Sustainable Development Goal 3**, making healthcare understanding accessible, inclusive, and actionable."

→ [Cue: Transition to Slide 8]

Slide 8: The Future Roadmap & Conclusion

🕒 0:30 mins

Stage Direction: Conclude firmly. Reiterate that the application is fully live and operational.

"Looking ahead, our roadmap includes **multilingual localization** in Hindi, Bengali, and Tamil to reach rural populations, integration with **Apple HealthKit and Google Health Connect**, and a **Clinician QR Portal**.

To conclude: *WellBridge AI doesn't replace doctors — it empowers patients to walk into clinic visits prepared, informed, and confident.*

The project is fully deployed and live at **wellbridgeai.resence.in**.

Thank you, respected evaluators. I am now open to your questions."



Viva Q&A Cheat Sheet: Top 5 Evaluator Questions & Answers

Likely Evaluator Question	Winning Technical Answer for Aishwarya
1. How do you handle OCR on blurry or noisy lab reports?	"We pass raw image bytes directly to Google Gemini's multimodal vision model with a strict JSON schema prompt that validates clinical units (e.g. g/dL, mg/dL) and provides graceful fallback error notifications if numbers cannot be resolved."
2. What if Gemini API goes down or exceeds rate limits?	"We engineered a 4-tier dynamic fallback model ladder in server.ts (gemini-3.5-flash-lite → 3.5-flash → 3.6-flash → 3.7-flash) that automatically catches API errors and attempts the next available model before returning a response."
3. Is WellBridge AI attempting to replace licensed doctors?	"No, sir/ma'am. WellBridge AI is strictly an educational and communication bridge. It features a mandatory medical disclaimer banner, does not prescribe drugs, and explicitly formulates questions for the patient to ask their physician."
4. How is patient medical data protected from unauthorized access?	"All patient data is isolated under per-user subcollections in Google Firebase Firestore with strict authentication rules (<code>request.auth.uid == userId</code>), guaranteeing zero cross-patient data exposure."
5. What is the Print Isolation Engine in your Doctor Brief?	"We wrote custom CSS print media queries (<code>@media print</code>) and container isolation rules that strip away web navigation headers, footers, and sidebars, rendering only the clean, 1-page clinical summary ready for physical printing."